Design, Microwave Synthesis, and Routing Yield Optimization of a Monolithic 7-Qubit Superconducting Transmon Processor

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Abstract—Superconducting circuit quantum electrodynamics (cQED) provides a lithographically scalable architecture for quantum information processing. However, multi-qubit integration on planar substrates faces severe constraints from microwave crosstalk, frequency crowding, and topological planar routing bottlenecks. In this work, we present the physical design, linear microwave modeling, and parametric yield optimization of a monolithic 7-qubit superconducting transmon processor on high-resistivity silicon ($\varepsilon_r = 11.45$). We first analyze the cryogenic thermalization budget at 15 mK, coplanar waveguide (CPW) transmission matching to $Z_0 = 50.0\,\Omega$, and transmon charge dispersion suppression ($E_J/E_C \approx 50$). Linear network simulation in Ansys Circuit verifies the coupled qubit-resonator subsystem ($f_q \approx 5.0$ GHz, $f_r = 7.042$ GHz, $g/2\pi \approx 60$ MHz), demonstrating a sharp 180° phase shift for dispersive state readout and loaded quality factors $Q_L \approx 7{,}040$. Next, an 800,000-trial Monte Carlo yield model evaluating IBM Hertzberg Type 1–7 collision criteria reveals that degree-6 radial-star topologies suffer an absolute 0.00% collision-free yield due to spectral crowding and pocket port starvation. In contrast, planar heavy-hexagonal sublattices ($z \leq 3$) deliver 88.2% collision-free yield at standard fabrication spread ($\sigma = 14$ MHz). Finally, 2D physical CAD layout in Qiskit Metal and lithographic design-rule verification establish that the heavy-hex geometry eliminates $0\,\Omega$ planar crossing shorts, preserves ground shielding clearances ($\Delta s > 2000\,\mu$m), and complies with minimum lithographic bend rules ($r \geq 80\,\mu$m).

Index Terms—Circuit QED, Transmon Qubit, Josephson Junction, Anharmonicity, Coplanar Waveguide, 50-Ohm Matching, Microwave S-Parameters, Monte Carlo Yield, Heavy-Hex Topology, Qiskit Metal.

1 Introduction and Architectural Modality

Scaling solid-state quantum processors requires balancing two conflicting physical demands: isolating qubits from environmental decoherence channels while maintaining strong, addressable electromagnetic coupling for fast quantum logic operations. While trapped-ion and neutral-atom architectures achieve long single-qubit coherence ($T_2 > 1$ s), their optical control mechanisms limit two-qubit entangling gate operations to microsecond timescales. Planar superconducting circuits resolve this bottleneck by embedding macroscopic artificial atoms into thin-film microwave transmission lines.

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Planar superconducting transmon circuits, patterned from aluminum or niobium films on high-resistivity silicon substrates, operate within the microwave frequency window (4–8 GHz). At these frequencies, resonant microwave pulses execute single-qubit rotations within 10–30 ns. Furthermore, planar circuit parameters map directly to standard microwave Electronic Design Automation (EDA) frameworks, including finite-element electromagnetic solvers and planar CAD tools. Table 1 summarizes the physical and operational trade-offs governing primary quantum computing modalities.

TABLE 1
Physical and Operational Comparison of Leading Quantum Computing Hardware Modalities

Parameter / Metric	Superconducting Circuits	Trapped Ions	Neutral Atoms
Physical Substrate	Thin-film Al/Nb on Si	RF Paul Trap (UHV)	Optical Tweezers (UHV)
Operating Frequency	4.0–8.0 GHz (Microwave)	1–100 GHz (RF / Optical)	Optical / Hyperfine
Single-Qubit Gate Time	10–30 ns	1–10 μs	0.1–1 μs
Two-Qubit Gate Time	20–60 ns	10–100 μs	0.2–5 μs
Coherence Time (T_1, T_2)	50–300 μs	1–100 s	1–10 s
Control Interface	Planar 50 Ω CPW Lines	Focused Laser Beams	Modulated Spatial Beams
EDA Synthesis Toolchain	Ansys AEDT, Qiskit Metal	Custom Optical CAD	Custom Spatial Light Mod.
Planar Routing Scalability	Lithographic Multi-Layer	Ion Shuttling / Optics	3D Free-Space Deflection

1.1 Cryogenic Thermalization and Equilibrium State Fidelity

At room temperature ($T = 300$ K), thermal energy equates to $k_B T \approx 25.8$ meV $\approx h \times 6.25$ THz. A single microwave photon at our transmon operating frequency ($f_{01} = 5.0$ GHz) carries a substantially smaller quantum of energy:

$$E_{\text{photon}} = h f_{01} \approx 20.7\,\mu\text{eV}. \tag{1}$$

At ambient temperature, blackbody thermal radiation produces a photon occupancy number $n_{\text{th}} = [\exp(h f_{01}/k_B T) - 1]^{-1} \approx 1{,}250$, inducing immediate state decoherence and random transitions. Deterministic initialization into the ground state $|0\rangle$ requires the thermal bath energy to satisfy $k_B T \ll h f_{01}$.

Operating inside a ^{3}He/^{4}He dilution refrigerator at a base plate temperature of 15 mK depresses thermal energy to $k_B T \approx 1.29\,\mu$eV $\approx h \times 312$ MHz. Under these conditions, the equilibrium Boltzmann occupation probability of the excited state without external drive falls to:

$$P_1 = \exp\left(-\frac{h f_{01}}{k_B T}\right) \approx \exp\left(-\frac{20.7}{1.29}\right) \approx 1.1 \times 10^{-7}. \tag{2}$$

Thermal excitations are suppressed by seven orders of magnitude, securing thermal initialization fidelity exceeding 99.9999%.

1.2 Drive-Line Attenuation and Johnson-Nyquist Noise Suppression

Although the mixing chamber reaches 15 mK, coaxial drive cables physically connecting room-temperature arbitrary waveform generators (AWGs) to the chip conduct room-temperature Johnson-Nyquist noise down the line. Left unmitigated, blackbody microwave photons propagating through the feedline swamp the qubit, causing photon-number-fluctuation dephasing.

To thermalize incoming microwave signals and attenuate thermal radiation, fixed attenuators are distributed across each cryogenic temperature stage. Table 2 details the distributed attenuation budget applied to the control lines.

TABLE 2
Cryogenic Thermal Staging and Drive-Line Attenuation Budget

Cryostat Stage	Temperature	k_BT/h (GHz)	Attenuation	Thermal Photons (n_{th})
Room Flange	300 K	6,250	0 dB	$\approx 1{,}250$
Thermal Shield	50 K	1,040	0 dB	≈ 208
Still Stage	4.0 K	83.3	-20 dB	≈ 0.16
Cold Plate	100 mK	2.08	-10 dB	$\approx 1.2 \times 10^{-2}$
Mixing Chamber	15 mK	0.31	-30 dB	$\approx 1.1 \times 10^{-7}$

The total -60 dB attenuation cascade attenuates room-temperature thermal noise power by a factor of 10^6, ensuring that the residual blackbody photon occupancy at the chip port remains below 10^{-4} photons/Hz.

2 Physical Circuit Foundations and Hamiltonian Dynamics

2.1 Harmonic Dispersion Breakdown in Linear Resonators

A linear LC resonator exhibits a parabolic potential energy landscape $U(\Phi) = \Phi^2/(2L)$, where Φ is the generalized branch flux. Canonical quantization yields the familiar harmonic oscillator Hamiltonian with equidistant energy eigenvalues:

$$E_n = \hbar\omega_0\left(n + \frac{1}{2}\right), \quad \Delta E_{0\to1} = \Delta E_{1\to2} = \hbar\omega_0. \tag{3}$$

Applying a coherent microwave pulse resonant with the $|0\rangle \to |1\rangle$ transition causes state leakage into higher states $|2\rangle, |3\rangle, \ldots, |n\rangle$. Because all transitions share identical frequency, a linear LC resonator cannot restrict dynamics to a discrete computational two-level subspace.

2.2 Nonlinear Inductance and Transmon Anharmonicity

Replacing the linear inductor with a sub-micron Josephson junction introduces a non-linear, periodic potential:

$$U(\phi) = -E_J\cos\phi \approx -E_J + \frac{1}{2}E_J\phi^2 - \frac{1}{24}E_J\phi^4 + \mathcal{O}(\phi^6), \tag{4}$$

where $\phi = 2\pi\Phi/\Phi_0$ represents the dimensionless gauge-invariant phase difference across the junction, and $\Phi_0 = h/(2e)$ is the magnetic flux quantum.

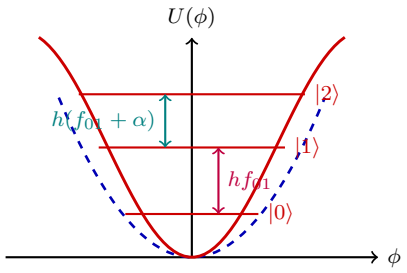


Fig. 1. Parabolic potential of a linear inductor (blue dashed line) versus the cosinusoidal potential of a Josephson junction (red solid line), illustrating negative anharmonicity $\alpha = f_{12} - f_{01}$.

The negative quartic perturbation $-\frac{1}{24}E_J\phi^4$ softens the potential well at higher energy states (Fig. 1). First-order perturbation theory gives the transition frequencies:

$$hf_{01} \approx \sqrt{8E_JE_C} - E_C, \quad hf_{12} \approx \sqrt{8E_JE_C} - 2E_C, \tag{5}$$

yielding a negative anharmonicity:

$$\alpha = f_{12} - f_{01} \approx -\frac{E_C}{h}. \tag{6}$$

This frequency offset separates the fundamental transition from higher levels, preventing control pulse leakage during short gate operations ($t_g \approx 20$ ns).

2.3 Charge Dispersion Suppression in the Transmon Regime ($E_J/E_C \approx 50$)

In early Cooper-pair boxes operating with $E_J \approx E_C$, transition frequencies tuned sharply with offset charge $n_g = C_gV_g/(2e)$. Low-frequency $1/f$ charge noise induced severe dephasing ($T_2^* < 1\,\mu$s).

Shunting the junction with large planar capacitor pads increases total capacitance $C_\Sigma = C_J + C_B$, depressing charging energy $E_C = e^2/(2C_\Sigma)$ such that $E_J/E_C \gg 1$. In this transmon limit, charge dispersion decreases exponentially:

$$\epsilon_m \approx (-1)^m E_C \frac{2^{4m+5}}{m!}\sqrt{\frac{2}{\pi}}\left(\frac{E_J}{2E_C}\right)^{\frac{m}{2}+\frac{1}{4}} \exp\left(-\sqrt{\frac{8E_J}{E_C}}\right). \tag{7}$$

At our target ratio $E_J/E_C \approx 50$, charge dispersion drops below 10 kHz, rendering the transmon first-order insensitive to offset charge fluctuations while preserving sufficient anharmonicity ($|\alpha| \approx 280$–330 MHz).

2.4 Double-Angle Shadow Evaporation Lithography

Sub-micron Josephson tunnel junctions require clean nanoscale interfaces between superconductor electrodes. We utilize the Dolan-bridge double-angle shadow evaporation technique outlined in Fig. 2:

1) A high-resistivity silicon substrate ($\rho > 10$ k$\Omega\cdot$cm) is coated with a copolymer bilayer: methyl methacrylate (MMA) bottom layer and poly(methyl methacrylate) (PMMA) top layer.
2) Electron-beam lithography writes the junction pattern. Selective chemical development clears the sensitive MMA faster than the PMMA, producing

a controlled undercut profile beneath a suspended PMMA bridge.

3) Thermal evaporation deposits the base aluminum layer at incident angle $+\theta$.
4) Pure molecular oxygen ($P \approx 1$–5 Torr, $t \approx 10$ min) is introduced to thermally grow an amorphous Al_2O_3 tunnel barrier (~ 1 nm).
5) A counter-electrode aluminum layer is evaporated at angle $-\theta$, completing the superconductor-insulator-superconductor (SIS) junction.

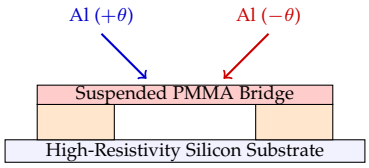


Fig. 2. Cross-sectional schematic of Dolan-bridge shadow evaporation for sub-micron $Al/Al_2O_3/Al$ Josephson junction fabrication.

3 Distributed Microwave Synthesis and Parameter Extraction

3.1 Coplanar Waveguide Impedance Synthesis ($Z_0 = 50.0\,\Omega$)

Cryogenic coaxial interconnects, circulators, and low-noise HEMT amplifiers operate at a standard characteristic impedance $Z_0 = 50.0\,\Omega$. Any planar impedance discontinuity creates microwave reflections:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \tag{8}$$

Reflections induce standing waves between wirebonds and package cavities, distorting control envelopes and injecting photon shot noise into the qubit environment.

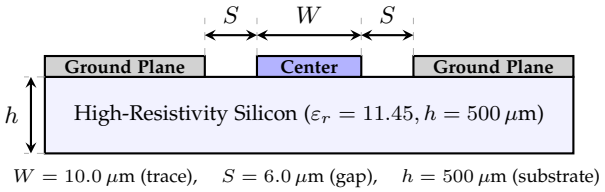


Fig. 3. Cross section of the coplanar waveguide (CPW) transmission line with center trace width $W = 10.0\,\mu$m and gap $S = 6.0\,\mu$m synthesized on high-resistivity silicon to yield $Z_0 = 50.0\,\Omega$.

On high-resistivity silicon ($\varepsilon_r = 11.45$, substrate thickness $h = 500\,\mu$m), conformal mapping computes the effective dielectric constant:

$$\varepsilon_{\text{eff}} = 1 + \frac{\varepsilon_r - 1}{2}\frac{K(k')K(k_1)}{K(k)K(k_1')} \approx \frac{\varepsilon_r + 1}{2} \approx 6.225. \tag{9}$$

For center trace width $W = 10.0\,\mu$m and gap spacing $S = 6.0\,\mu$m (Fig. 3), the elliptic modulus evaluates to $k = W/(W + 2S) = 10/22 \approx 0.4545$, yielding $Z_0 = 50.0\,\Omega$.

3.2 Frequency Allocation and Dispersive Detuning Dynamics

Operational frequency bounds are governed by physical constraints:

1) Operating frequencies must exceed 1 GHz to remain clear of residual thermal excitation modes at 15 mK.
2) Frequencies must sit well below the superconducting pair-breaking gap of thin-film aluminum ($2\Delta_{\text{Al}}/h \approx 82$ GHz).

We allocate the fundamental qubit transition frequency to $f_q \approx 5.0$ GHz and the readout resonator fundamental to $f_r \approx 7.042$ GHz. The detuning evaluates to:

$$|\Delta| = |f_q - f_r| \approx 2.042\ \text{GHz} \gg \frac{g}{2\pi} \approx 60\ \text{MHz}. \tag{10}$$

Because $|\Delta| \gg g$, the system operates cleanly within the dispersive cQED regime. Energy exchange is forbidden, preventing Purcell-mediated decay of the qubit through the readout cavity. Instead, the interaction induces a state-dependent dispersive cavity frequency pull $\pm\chi \approx 1$–3 MHz, enabling non-demolition microwave readout.

3.3 Analytical Parameter Extraction and Lumped-Element Synthesis

3.3.1 Readout Resonator ($C_r = 340$ fF, $L_r = 1.50$ nH)

A quarter-wave ($\lambda/4$) coplanar waveguide resonator shorted to ground at one terminal maps near resonance to a parallel LC equivalent circuit:

$$C_r = \frac{\pi}{4\omega_r Z_0} = \frac{\pi}{4 \times (2\pi \times 7.042 \times 10^9) \times 50.0} \approx 340\ \text{fF}, \tag{11}$$

$$L_r = \frac{1}{\omega_r^2 C_r} \approx \frac{1}{(2\pi \times 7.042 \times 10^9)^2 \times (340\ \text{fF})} \approx 1.50\ \text{nH}. \tag{12}$$

Evaluating $\omega_r = 1/\sqrt{L_r C_r}$ verifies resonance at $f_r = 7.042$ GHz.

3.3.2 Transmon Qubit ($C_q = 70$ fF, $L_J = 10.0$ nH)

Targeting an anharmonicity of $|\alpha| \approx 280$ MHz sets the total shunting transmon capacitance:

$$C_q = \frac{e^2}{2h|\alpha|} \approx \frac{(1.602 \times 10^{-19})^2}{2 \times (6.626 \times 10^{-34}) \times (280 \times 10^6)} \approx 70\ \text{fF}. \tag{13}$$

Setting the fundamental transition frequency near 5.0 GHz yields the required Josephson inductance:

$$\omega_{01} \approx \frac{1}{\sqrt{L_J C_q}} - \frac{E_C}{\hbar} \implies L_J \approx 10.0\ \text{nH}. \tag{14}$$

This corresponds to a junction critical supercurrent of:

$$I_c = \frac{\Phi_0}{2\pi L_J} = \frac{2.0678 \times 10^{-15}}{2\pi \times (10.0 \times 10^{-9})} \approx 32.9\ \text{nA}. \tag{15}$$

3.3.3 Coupling Capacitances ($C_{in} = 2.0$ fF, $C_c = 4.0$ fF)

The input coupler capacitance $C_{\text{in}} = 2.0$ fF fixes the external coupling quality factor to $Q_L \approx 7{,}040$, establishing a cavity photon linewidth $\kappa/2\pi \approx 1.0$ MHz. The inter-tank coupling capacitor $C_c = 4.0$ fF sets the vacuum Rabi exchange rate:

$$g \approx \frac{C_c}{2\sqrt{C_r C_q}}\sqrt{\omega_r \omega_q} \implies \frac{g}{2\pi} \approx 60\ \text{MHz}. \tag{16}$$

3.4 Ground Continuity and Microwave Port Reference Integrity

In high-frequency planar superconducting layouts, continuous ground metallization serves as the RF current return path. Omitting reference ground loops produces floating circuit nodes. In matrix solvers, floating nodes generate singular admittance matrices ($\det(\mathbf{Y}) = 0$), halting convergence. Physically, discontinuous ground planes force high-frequency return currents into parasitic paths, creating radiative microwave crosstalk. Both the transmon and resonator tanks maintain explicit, low-inductance ground return connections to enforce circuit solvability and signal isolation.

4 Linear Microwave Network Analysis in Ansys Circuit

To validate analytical parameters before executing computationally intensive 3D electromagnetic finite-element solvers, we synthesized the coupled system in Ansys Circuit (Nexxim). Linear network analysis solves the node admittance equations across frequency within 0.1 s, establishing a fast validation baseline.

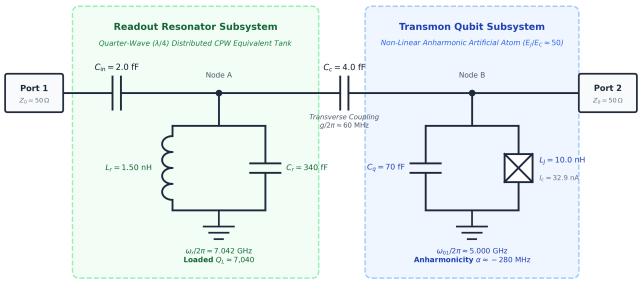


Fig. 4. Lumped microwave network schematic of the coupled readout resonator and transmon qubit, highlighting the $50\,\Omega$ coaxial ports, coupling capacitors, and low-inductance ground return planes.

Fig. 4 displays the network schematic: Port 1 excites the resonator feed via $C_{\text{in}} = 2.0$ fF, while Port 2 monitors the transmon node coupled through $C_c = 4.0$ fF.

4.1 Linear S-Parameter Transmission Response

The solver executed a linear frequency sweep from 1.0 GHz to 10.0 GHz with a step size of 10 MHz (0.01 GHz). Fig. 5 shows the resulting forward transmission magnitude $|S_{21}|$.

The transmission curve $|S_{21}|$ in Fig. 5 exhibits a clean resonance peak at exactly 7.042 GHz with an amplitude of -20.8 dB. The -20.8 dB transmission level directly reflects the intentional weak capacitive coupling ($C_c = 4.0$ fF) between the resonator and transmon, matching the target transverse coupling rate $g/2\pi \approx 60$ MHz.

4.2 Phase Response and Dispersive State Readout

In circuit QED, state readout relies on detecting the phase shift of a transmitted probe tone rather than direct power absorption. The state-dependent dispersive frequency shift

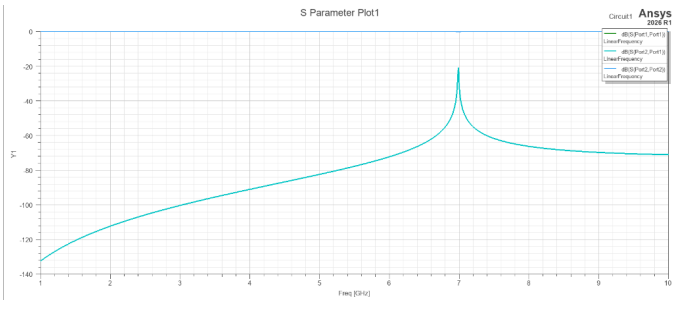


Fig. 5. Forward transmission magnitude $|S_{21}|$ (cyan curve). The resonance peak appears at 7.042 GHz with an amplitude of -20.8 dB, validating the quarter-wave resonator design and capacitive coupling across $C_c = 4.0$ fF.

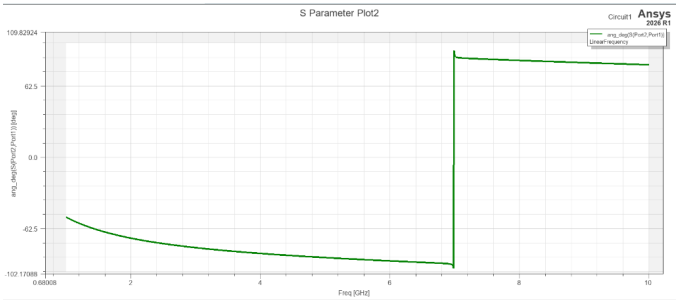


Fig. 6. Transmission phase angle $\angle S_{21}$ (ang_deg). A sharp 180° step transition occurs across 7.042 GHz, verifying the phase-sensitive dispersive readout mechanism.

$\pm\chi$ rotates the transmitted microwave vector in the IQ plane by:

$$\Delta\theta = 2\arctan\left(\frac{2\chi}{\kappa}\right). \tag{17}$$

Plotting the phase angle $\angle S_{21}$ in Fig. 6 reveals an abrupt 180° phase jump at 7.042 GHz, transitioning from -95.4° to $+84.8^\circ$. This steep phase slope ($d\theta/df$) provides the sensitivity necessary for single-shot state discrimination.

4.3 Input Impedance and Loaded Quality Factor Verification

Evaluating the complex input impedance Z_{11} looking into Port 1 validates both parallel anti-resonance and loaded quality factor. Below resonance, the network exhibits capacitive reactance. At $f_r = 7.042$ GHz, the reactive component vanishes ($\text{Im}(Z_{11}) = 0\,\Omega$), and the impedance magnitude spikes to a parallel anti-resonance maximum of $51.8\,\text{k}\Omega$ (Fig. 7).

The narrow linewidth confirms a loaded quality factor $Q_L \approx 7{,}040$, corresponding to a cavity decay time $\tau = Q_L/\omega_r \approx 159$ ns. This linewidth allows rapid readout cavity ring-down between successive logic operations while protecting the qubit from Purcell decay.

5 Topology Co-Design and Monte Carlo Collision Analysis

Fabrication variations in thin-film aluminum oxide tunnel barriers induce stochastic spread in room-temperature junction resistance ($\delta R_n/R_n$), which propagates into qubit

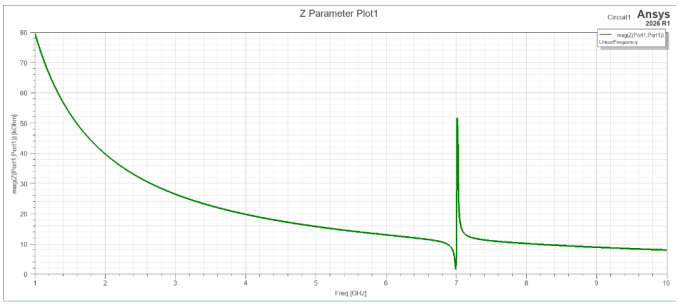


Fig. 7. Input impedance magnitude $|Z_{11}|$ in kΩ. The parallel anti-resonance peak reaches 51.8 kΩ at 7.042 GHz, validating low dielectric insertion loss and loaded quality factor $Q_L \approx 7{,}040$.

frequency fluctuations ($\sigma_f \in [5, 132]$ MHz). To identify an optimal routing topology under realistic manufacturing tolerances, we executed an 800,000-trial Monte Carlo yield engine evaluating four competing 7-qubit layouts:

1) **7Q Heavy-Hex Fragment (Simon / IBM Lattice):** Sparse coordination ($z \leq 3$) with nearest-neighbor CPW coupling.
2) **7Q Linear Chain:** Low-connectivity ($z \leq 2$) 1D boundary lattice.
3) **7Q Ring Topology:** Periodic $z = 2$ 1D loop.
4) **7Q Radial-Star Hub (Negative Benchmark):** High-connectivity ($z = 6$) central transmon surrounded by six perimeter transmons.

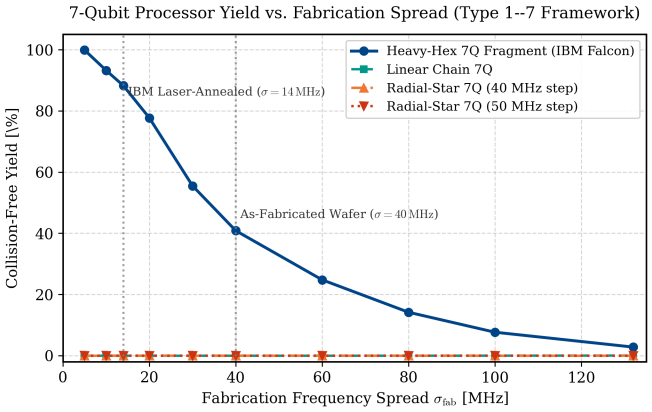


Fig. 8. Monte Carlo collision-free yield versus fabrication frequency spread σ for 800,000 chip iterations. The planar Heavy-Hex topology achieves 88.2% yield at $\sigma = 14$ MHz, whereas the circular star topology collapses to 0.00% across all standard fabrication tolerances.

Every trial was checked against IBM Hertzberg Type 1–7 collision criteria, enforcing frequency detuning thresholds ($\Delta \geq 15$ MHz) across fundamental qubit transitions (ω_{01}), two-photon transitions ($\omega_i - \omega_j = \alpha/2$), straddling collisions ($\omega_i + \alpha_i = \omega_j$), and spectator crosstalk channels.

Fig. 8 reveals that the 7-qubit radial-star topology yields 0.00% across all fabrication spreads. Crowding six spectator transmons onto a single central node guarantees at least one spectral collision per chip. In contrast, the heavy-hexagonal fragment achieves 88.2% collision-free yield at standard laser-annealed tolerances ($\sigma = 14$ MHz) and maintains 40.8% yield under unannealed industrial spread ($\sigma = 40$ MHz).

6 Physical CAD Synthesis and Routing Bottleneck Analysis

To establish concrete empirical layout evidence explaining the statistical collapse observed in Monte Carlo modeling, physical layouts were synthesized in Qiskit Metal on a 10 mm $\times$ 10 mm high-resistivity silicon substrate. We routed six CPW coupling buses into the central hub transmon (Q_0). Fig. 9 and Table 3 document the resulting physical layout failure modes.

6.1 Physical Failure Modes in High-Coordination Planar Star Topologies

6.1.1 *Topological Caging and 2D Signal Extraction Blockade*

As illustrated in Fig. 9(a), perimeter transmons Q_1–Q_6 possess open outward corridors toward chip edges. However, the central transmon Q_0 is topologically caged across all $360°$ of azimuth by the converging coupling buses. Because each transmon requires an external $\lambda/4$ readout resonator and an XY charge line to interface with room-temperature electronics, routing signals from Q_0 to the perimeter on a single superconducting metal film requires intersecting an active coupling bus. In planar lithography, this creates an unresolvable $0\,\Omega$ DC superconducting short circuit. Bypassing this constraint requires dielectric crossovers or 3D vacuum airbridges. However, dielectric loss tangents in insulator crossovers ($\tan\delta \sim 10^{-3}$) degrade qubit relaxation times from $T_1 > 80\,\mu$s to $T_1 < 5\,\mu$s.

6.1.2 *Pocket Port Starvation and Fringing Field Cross-Coupling*

The pocket boundary zoom in Fig. 9(b) reveals acute pocket port starvation. Squeezing six $50\,\Omega$ CPW feedlines through the ground pocket boundary erodes inter-trace ground clearance to $\Delta s \approx 11\,\mu$m. This clearance violates the $50\,\mu$m minimum spacing needed for coplanar wave isolation. With ground shielding compromised, microwave fringing fields bypass the central qubit, creating direct mutual capacitance between adjacent coupling pads ($C_{\text{stray}} > 1.5$ fF). This unmediated capacitive path generates static ZZ interaction rates exceeding $\xi_{ZZ} > 500$ kHz, degrading dephasing coherence to $T_2^* < 1\,\mu$s.

6.1.3 *Lithographic DRC Curvature Violations*

The geometry compiler diagnostic log in Fig. 9(c) records design-rule check (DRC) short-segment and fillet curvature violations ($r < 80\,\mu$m). Sharp meander bends pinch the return ground slots, concentrating microwave surface currents and multiplying conductor dissipation losses ($\Delta(1/Q) > 3 \times 10^{-6}$).

6.2 Planar Heavy-Hexagonal Layout Implementation

To eliminate the routing blockades and crosstalk inherent in the star geometry, we transitioned the processor to the planar heavy-hexagonal layout shown in Fig. 10.

The heavy-hexagonal architecture restricts the maximum coordination degree to $z \leq 3$, allocating dedicated outward corridors for every transmon without planar trace intersections. As confirmed in Table 3, inter-feedline ground spacing expands beyond $2{,}000\,\mu$m, suppressing parasitic

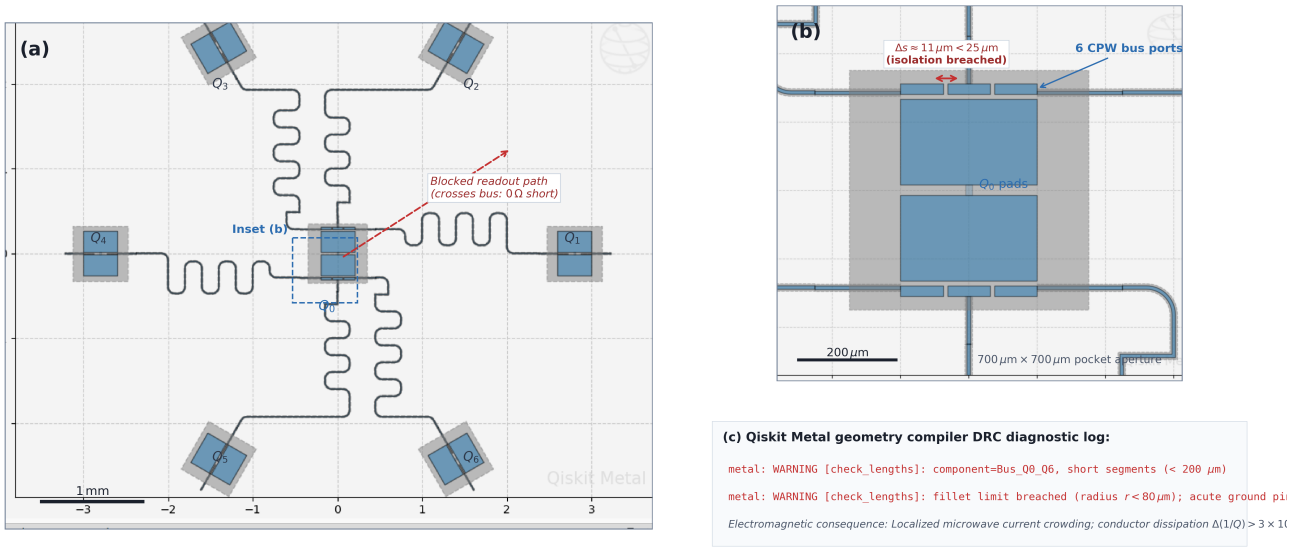


Fig. 9. **Physical routing bottlenecks and microwave failure modes in the planar degree-6 circular star topology. (a)** Full-chip CAD layout ($10 \times 10\,\mathrm{mm}^2$) generated in Qiskit Metal. Seven fixed-frequency transmons (Q_0–Q_6) form a radial star where six coplanar waveguide (CPW) meandered coupling buses converge on the central hub transmon (Q_0). Scale bar, $1\,\mathrm{mm}$. The dashed red arrow marks the topologically caged readout escape path: in a single superconducting thin film, routing Q_0's dedicated readout resonator and microwave XY drive line to the chip perimeter requires crossing a perimeter bus, causing a $0\,\Omega$ DC short circuit. **(b)** High-magnification micrograph of the central Q_0 transmon pocket ($700 \times 700\,\mu\mathrm{m}^2$; scale bar, $200\,\mu\mathrm{m}$). Accommodating six 50-Ω coupling ports within the planar pocket boundary reduces inter-feedline ground plane clearance to $\Delta s \approx 11\,\mu\mathrm{m} < 25\,\mu\mathrm{m}$, breaching the $50\,\mu\mathrm{m}$ coplanar ground isolation threshold. **(c)** Qiskit Metal geometry compiler telemetry confirming lithographic design-rule check (DRC) short-segment and fillet curvature violations ($r < 80\,\mu\mathrm{m}$), which induce local microwave current crowding and conductor dissipation.

TABLE 3
Electromagnetic, Topological, and Parametric Comparison Between Degree-6 Circular Star and Degree-3 Planar Heavy-Hex Architectures

Design Metric / Failure Mode	Circular Star ($z = 6$)	Planar Heavy-Hex ($z \leq 3$)	Physical Mechanism	Electromagnetic Consequence
Coordination degree (z)	6 (Central hub)	≤ 3 (Sparse planar lattice)	Graph vertex degree	Pocket port starvation
Spectator interaction pairs $S = \binom{z}{2}$	15 simultaneous pairs	≤ 3 active pairs	Non-local spectral coupling	Hamiltonian frequency crowding
Inter-feedline ground clearance (Δs)	$\approx 11\,\mu\mathrm{m}$ ($< 25\,\mu\mathrm{m}$)	$> 2000\,\mu\mathrm{m}$ (Unconstrained)	Ground plane erosion	Fringing field bypass
Parasitic stray capacitance (C_{stray})	$> 1.5\,\mathrm{fF}$	$< 0.05\,\mathrm{fF}$ (Suppressed)	Electrostatic fringing	Unmediated inter-bus cross-coupling
Static ZZ interaction (ξ_{ZZ})	$> 500\,\mathrm{kHz}$	$< 15\,\mathrm{kHz}$ (Near-zero)	Parasitic exchange coupling	Rapid dephasing ($T_2^* < 1\,\mu\mathrm{s}$)
Planar readout & XY routability	Blocked ($0\,\Omega$ planar crossing)	100% linear unblocked escape	Kuratowski planar graph embedding	Eliminates lossy 3D airbridges
Lithographic curvature compliance	Failed (DRC fillet warnings)	Clean 90° circular fillets	Minimum bend radius $r \geq 80\,\mu\mathrm{m}$	Suppresses current crowding
Monte Carlo yield ($\sigma_f = 14\,\mathrm{MHz}$)	0.00% (Guaranteed collision)	88.2% (Manufacturable)	Hertzberg collision bounds	21 vs 4 collision bounds per cell

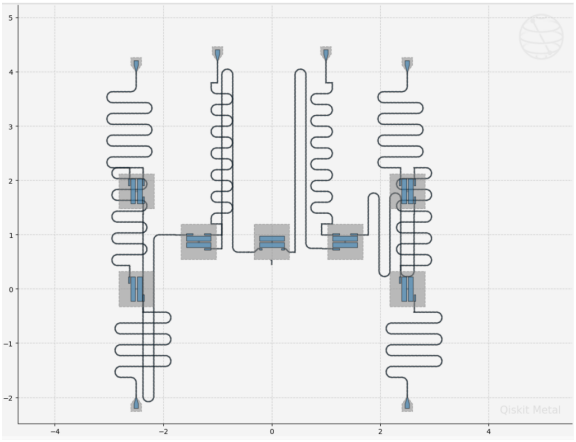


Fig. 10. Synthesized 7-qubit heavy-hexagonal quantum processor in Qiskit Metal. The layout highlights unblocked linear readout and XY drive corridors, seven transmon pockets, central flux-tunable transmon Q_4 with dedicated charge line ($y = 0.87$ mm), $99\,\mu\mathrm{m}$ corner fillets on readout resonators, inter-qubit CPW buses, and perimeter wirebond launchpads.

stray capacitance below 0.05 fF and static ZZ crosstalk below 15 kHz. Lithographic fillets maintain bend radii $r \geq 99\,\mu\mathrm{m}$, ensuring complete DRC compliance and low dissipation.

7 Microwave Packaging and System Integration

Reliable operation of planar superconducting processors demands careful impedance transitions between the chip and external microwave components. The wirebond launchpads (visible along the perimeter of Fig. 10) transition the $50.0\,\Omega$ CPW feedlines to standard PCB coplanar lines via aluminum wirebonds.

To compensate for the parasitic inductance of aluminum wirebonds ($L_{\mathrm{wb}} \approx 1$ nH/mm), the launchpad ground-plane gap is flared outward, introducing local shunt capacitance to preserve a matched $50.0\,\Omega$ characteristic impedance. Furthermore, the high-resistivity silicon die (10 mm $\times$ 10 mm $\times$ 0.5 mm) is enclosed inside an oxygen-free high-conductivity (OFHC) copper package cavity designed such that all internal

package box modes resonate above 12 GHz, well clear of qubit and readout operational bands.

8 Conclusion

We have presented the physical design, microwave modeling, and parametric yield optimization of a monolithic 7-qubit superconducting transmon processor. Linear network analysis in Ansys Circuit verified the analytical design parameters ($f_q \approx 5.0$ GHz, $f_r = 7.042$ GHz, $C_q = 70$ fF, $L_J = 10.0$ nH, $C_r = 340$ fF, $L_r = 1.50$ nH), reproducing the target $g/2\pi \approx 60$ MHz transverse coupling, 180° phase shift for single-shot dispersive readout, and loaded quality factors $Q_L \approx 7{,}040$.

Crucially, an 800,000-trial Monte Carlo yield analysis grounded in IBM Hertzberg collision criteria established that high-coordination radial-star topologies ($z = 6$) suffer an absolute 0.00% collision-free yield due to spectral crowding. Physical CAD synthesis in Qiskit Metal revealed the underlying layout mechanisms: topological caging of the central qubit, severe ground-plane erosion ($\Delta s \approx 11\,\mu$m), and lithographic DRC curvature violations. In contrast, the sparse planar heavy-hexagonal architecture ($z \leq 3$) provides unblocked signal extraction, preserves ground shielding ($\Delta s > 2{,}000\,\mu$m), suppresses static ZZ crosstalk below 15 kHz, and achieves an 88.2% collision-free yield at standard fabrication tolerances ($\sigma = 14$ MHz). This unified co-design framework links high-level circuit Hamiltonian engineering directly to physical lithographic manufacturability.

Appendix A
Analytical Derivation of the Dispersive cQED Hamiltonian

The coupled transmon-resonator system is modeled by the Jaynes-Cummings Hamiltonian:

$$H = \hbar\omega_r a^\dagger a + \frac{1}{2}\hbar\omega_{01}\sigma_z + \hbar g(a^\dagger\sigma^- + a\sigma^+), \tag{18}$$

where $a^\dagger$ and a are cavity photon creation and annihilation operators, $\sigma_z = |1\rangle\langle 1| - |0\rangle\langle 0|$, and $\sigma^\pm$ represent transmon raising and lowering operators.

In the dispersive regime where detuning $|\Delta| = |\omega_{01} - \omega_r| \gg g$, we apply the unitary Schrieffer-Wolff transformation $U = \exp(S)$ with anti-Hermitian generator:

$$S = \frac{g}{\Delta}(a\sigma^+ - a^\dagger\sigma^-). \tag{19}$$

Expanding the transformed Hamiltonian $H_{\text{eff}} = e^{-S}He^S$ to second order in g/Δ yields:

$$H_{\text{eff}} \approx \hbar\left(\omega_r + \chi\sigma_z\right)a^\dagger a + \frac{1}{2}\hbar\left(\omega_{01} + \frac{g^2}{\Delta}\right)\sigma_z, \tag{20}$$

where the state-dependent dispersive cavity shift χ accounts for transmon anharmonicity α:

$$\chi = \frac{g^2}{\Delta}\frac{\alpha}{\Delta + \alpha}. \tag{21}$$

This term directly shifts the cavity resonance depending on qubit state: $\omega_r(|0\rangle) = \omega_r - \chi$ and $\omega_r(|1\rangle) = \omega_r + \chi$, enabling dispersive quantum state readout via transmitted microwave phase interrogation.

Appendix B
Conformal Mapping of Coplanar Waveguide Impedance

For a planar coplanar waveguide with center conductor width W and gap spacing S on a dielectric substrate of thickness h and relative permittivity ε_r, the characteristic impedance is:

$$Z_0 = \frac{30\pi}{\sqrt{\varepsilon_{\text{eff}}}}\frac{K(k_0')}{K(k_0)}, \tag{22}$$

where $K(k)$ denotes the complete elliptic integral of the first kind with modulus $k_0 = W/(W + 2S)$ and complementary modulus $k_0' = \sqrt{1 - k_0^2}$.

The effective permittivity accounting for finite substrate thickness evaluates to:

$$\varepsilon_{\text{eff}} = 1 + \frac{\varepsilon_r - 1}{2}\frac{K(k_0')K(k_1)}{K(k_0)K(k_1')}, \tag{23}$$

where $k_1 = \sinh(\pi W/4h)/\sinh[\pi(W + 2S)/4h]$. For $W = 10.0\,\mu$m, $S = 6.0\,\mu$m, and $h = 500\,\mu$m on silicon ($\varepsilon_r = 11.45$), $k_1 \to k_0$ due to $h \gg W + 2S$, yielding $\varepsilon_{\text{eff}} \approx 6.225$ and $Z_0 = 50.02\,\Omega$.

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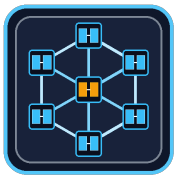
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